TEXORA · LECTURE SHEET

Project Management

Planning and controlling one-off undertakings — a new dyeing line, a factory extension, a compliance upgrade: life cycle, work breakdown, Gantt charts, and the network methods CPM and PERT that find the critical path and the true project duration.

TE-3109 · SUPPLY CHAIN MANAGEMENT IN TEXTILES · TOPIC 8 · LEVEL 3 TERM 1

1. What makes a project

A **project** is a temporary undertaking with a defined start and finish, created to produce a unique result. It differs from routine operations in four ways: it is **temporary** (the team disbands at the end), **unique** (no two factory installations are identical), **progressively elaborated** (detail is added as it proceeds), and **constrained** by scope, time, cost and quality — the four sides of the project constraint diamond, where squeezing one side stretches another. Textile examples are everywhere: installing a new dyeing machine line, building an effluent treatment plant (ETP) upgrade, preparing a factory for a social-compliance audit, launching a new product range, or relocating a sewing floor. Each needs project management precisely because it cannot be run by routine.

Project management is the application of knowledge, skills and techniques to meet the project's requirements: defining scope, planning time and cost, organising people, controlling progress and closing properly. The project manager's central problem is that the work is interdependent — tasks wait for other tasks — so overall duration cannot be found by adding task times. Network methods, covered in Sections 5 and 6, exist to solve exactly that problem.

2. The project life cycle

PHASE	MAIN WORK	KEY OUTPUT	TEXTILE EXAMPLE
1. Initiation	Define the need, objectives, stakeholders and feasibility; appoint the manager	Project charter / approval	Decision to install two new dyeing machines to cut lead time
2. Planning	Scope, work breakdown, schedule, budget, quality, risk and procurement plans	Project plan and baseline schedule	Machine specification, supplier selection, installation sequence

3. Execution	Do the work: procure, build, install, train; coordinate people and contractors	Deliverables taking shape	Foundation, machine erection, piping and wiring
4. Monitoring and control	Compare progress with plan; manage changes, risks and quality (runs alongside execution)	Progress reports, revised forecasts	Weekly checks against the installation schedule
5. Closure	Commission, hand over, document lessons, release the team	Accepted deliverable and closure report	Trial production accepted; line handed to production

Effort is low at initiation, peaks during execution and falls at closure, while the *cost of changes* moves the opposite way — a change of machine position costs almost nothing on paper during planning and a fortune after the foundation is cast. This single fact justifies heavy investment in careful planning. **Stakeholders** — owners, production managers, buyers, workers, regulators, neighbours — are identified at initiation because their requirements define success as much as the technical specification does.

3. Work breakdown structure and responsibility

The **work breakdown structure (WBS)** decomposes the project into manageable pieces: project, then major deliverables, then work packages small enough to estimate, assign and control. For a dyeing-line installation the first level might be: civil works; machine procurement; utilities; installation; commissioning. Each branch is broken down until a work package has one owner, a clear output and an estimable duration and cost. The WBS is the foundation of everything else — the schedule sequences WBS packages, the budget prices them, and the responsibility matrix assigns them.

A **Gantt chart** then displays the schedule as horizontal bars against a calendar: one row per activity, bar length equal to duration, position showing start and finish. Gantt charts are unmatched for communication — any manager can read one at a glance — and_dependencies can be shown with arrows. Their weakness is that in a large project the logic becomes hard to follow, which is why network analysis is done first and the Gantt chart drawn from its results. Milestones — zero-duration markers such as "machines delivered" or "trial run passed" — are plotted as diamonds on the chart.

4. Scheduling concepts: dependencies, float and the critical path

Activities are linked by **dependencies**, most commonly finish-to-start: the successor cannot start until the predecessor finishes (foundation before machine erection). For each activity the forward pass computes the **earliest start (ES)** and **earliest finish (EF = ES + duration)**; the backward pass

computes the **latest finish (LF)** and **latest start (LS = LF − duration)**. **Total float (slack)** = LS − ES = LF − EF: the time an activity can slip without delaying the project. The **critical path** is the longest path through the network; its activities have zero float, and the project duration equals its length. Any delay on the critical path delays the project day for day — so management attention, best resources and closest monitoring belong there. **Free float**, the slip possible without delaying any successor's earliest start, is always less than or equal to total float.

5. CPM — the critical path method

CPM is used when activity durations can be estimated with reasonable confidence from experience. Steps: list activities and predecessors; draw the network; estimate durations; forward pass for ES/EF; backward pass for LS/LF; compute floats; identify the critical path. The worked network below is the standard teaching example for this course and should be mastered in full.

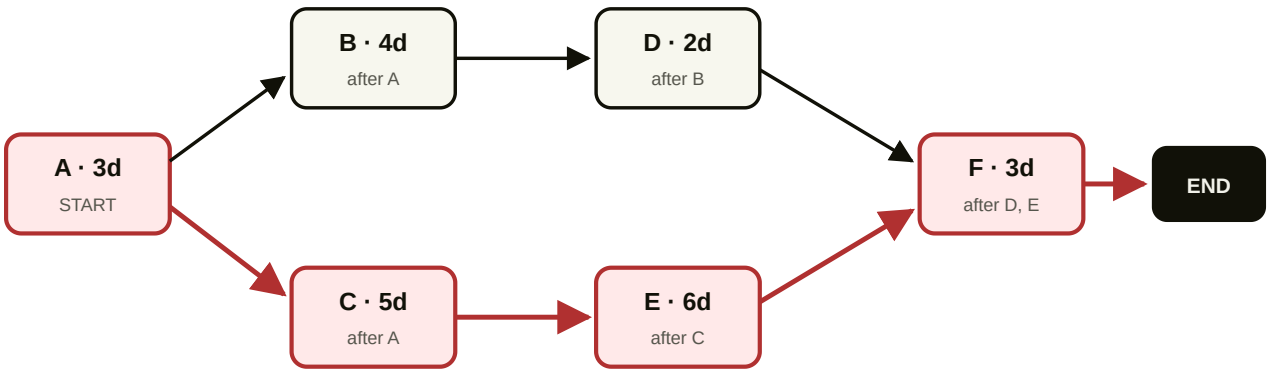


Figure 1 — Precedence network for the worked example. Activities A–F with durations in days. The critical path A–C–E–F (17 days) is highlighted; the parallel path A–B–D–F takes only 12 days and therefore carries 5 days of float.

WORKED EXAMPLE — CPM NETWORK

Activities: A (3 days) starts the project; B (4 days) follows A; C (5 days) follows A; D (2 days) follows B; E (6 days) follows C; F (3 days) follows both D and E.

Path 1: A–B–D–F = 3 + 4 + 2 + 3 = **12 days**.

Path 2: A–C–E–F = 3 + 5 + 6 + 3 = **17 days**.

The longer path is the **critical path: A–C–E–F**, and the **project duration is 17 days**. Path 1 has 17 − 12 = 5 days of total float shared among B and D, so B or D can slip modestly without consequence — but any slip in A, C, E or F delays the whole project.

Crashing is the deliberate shortening of the project by spending more on critical activities — overtime, extra crews, faster shipping. Only crashing a critical activity shortens the project, and as activities are crashed the critical path can change, so the network must be re-examined after each step. The cheapest critical activity (lowest cost slope, extra cost per day saved) is crashed first, and crashing stops when the cost of saving a day exceeds the benefit — for example a buyer's penalty clause or a bonus for early completion.

6. PERT — planning under uncertainty

PERT (Programme Evaluation and Review Technique) handles activities whose durations are uncertain — first-time work, imported machinery, approvals. Each activity gets three estimates: optimistic (o), most likely (m) and pessimistic (p). The expected duration and variance are:

$$t_e = (o + 4m + p) \div 6 \quad \cdot \quad \text{Variance} = [(p - o) \div 6]^2$$

The expected duration of the project is the sum of t_e along the (expected) critical path, and the project variance is the sum of the critical activities' variances — variances add, standard deviations do not. With the project standard deviation, the probability of meeting a deadline follows from the normal distribution using Z = (deadline − expected duration) ÷ SD.

WORKED EXAMPLE — PERT ON THE CRITICAL PATH

For the critical activities of the network above, three-point estimates are:

A (o 2, m 3, p 4): $t_e = (2 + 12 + 4) \div 6 =$ **3.0 days**, variance = $(2 \div 6)^2 =$ **0.111**.

C (o 3, m 5, p 7): $t_e = (3 + 20 + 7) \div 6 =$ **5.0 days**, variance = $(4 \div 6)^2 =$ **0.444**.

E (o 4, m 6, p 8): $t_e = (4 + 24 + 8) \div 6 =$ **6.0 days**, variance = $(4 \div 6)^2 =$ **0.444**.

F (o 2, m 3, p 4): $t_e = (2 + 12 + 4) \div 6 =$ **3.0 days**, variance = $(2 \div 6)^2 =$ **0.111**.

Expected project duration = 3.0 + 5.0 + 6.0 + 3.0 = **17 days** (matching CPM). Project variance = 0.111 + 0.444 + 0.444 + 0.111 = **1.111**, so SD = $\sqrt{1.111} \approx$ **1.05 days**.

Probability of finishing within 19 days: Z = (19 − 17) ÷ 1.05 ≈ **1.90**, which from the standard normal table gives a probability of about **97 %**. The deadline is therefore a sound promise — but a 17-day promise would be only a 50 % promise, because the expected duration is by definition the fifty-fifty point.

CPM and PERT are complements, not rivals. CPM suits repetitive, well-understood work with firm durations; PERT suits novel work with uncertain durations; modern practice merges them — a CPM network whose risky activities carry three-point estimates. Both rest on the same discipline: an honest activity list, honest predecessors and honest estimates. A network built to please management rather than to describe the work will fail precisely when it is needed most.

7. Cost control, risk and closure

Time and cost are controlled together. The planned budget is spread over the schedule to give a baseline; comparing work actually earned against money actually spent (the essence of **earned value** thinking) reveals whether the project is behind, ahead, over or under budget, while simple percentage-complete reports conceal all four combinations. **Risk management** identifies what might go wrong — machine shipment delay, power connection delay, monsoon disruption of civil work — assesses likelihood and impact, and prepares responses: avoid, reduce, transfer (insurance, contract terms) or accept with a contingency reserve. Closure then formalises acceptance, hands over documentation and spares, settles contracts, and records lessons learned so the next installation starts wiser. Skipping closure is why factories repeat the same installation mistakes machine after machine.

Glossary

Project — a temporary, unique undertaking with defined objectives and constraints. **Life cycle** — the phases from initiation to closure. **WBS** — work breakdown structure; hierarchical decomposition into work packages. **Gantt chart** — bar chart of activities against calendar time. **Critical path** — the longest network path; it sets the project duration and has zero float. **Float / slack** — time an activity can slip without delaying the project (total float) or its successors (free float). **CPM** — critical path method using single duration estimates. **PERT** — network method using three-point estimates and probabilities. **Crashing** — shortening critical activities at extra cost. **Milestone** — a significant zero-duration event in the schedule.

Key points

• A project is temporary and unique, constrained by scope, time, cost and quality; its life cycle runs initiation → planning → execution (with monitoring and control) → closure.

• The WBS decomposes the work; the Gantt chart communicates the schedule; network analysis determines it.

• The critical path is the longest path and sets the duration: in the worked network, A–C–E–F = 17 days is critical, while A–B–D–F = 12 days carries 5 days of float.

• Critical activities have zero float — delay there is project delay. Crash only critical activities, cheapest cost slope first.

• PERT: $t_e = (o + 4m + p)/6$; project variance is the sum of critical variances. Worked result: expected duration 17 days, variance 1.111, SD ≈ 1.05 days, probability of finishing within 19 days ≈ 97 % ($Z \approx 1.90$).

• Control time and cost together, manage risks before they occur, and close formally with lessons learned.

Practice

1. A network has paths P–Q–S = 4 + 6 + 3 days and P–R–S = 4 + 9 + 3 days. Which path is critical, what is the project duration, and how much float does the shorter path carry?
2. An activity has o = 5, m = 8, p = 17 days. Calculate its expected duration and variance.
3. Draw the Gantt-chart information for the worked network in this sheet: list each activity with its earliest start and earliest finish, taking day 0 as the project start.
4. A project's expected duration is 40 days with SD 2 days. Estimate the probability of finishing within 44 days, given that Z = 2.0 corresponds to about 97.7 %. Show the Z calculation.
5. Explain the difference between total float and free float, and why a project manager watches critical activities more closely than floating ones.
6. A dyeing-line installation is two weeks behind and the buyer-linked opening date cannot move. Using the idea of crashing, state the steps you would take and two risks of crashing.

Answer key

1. P–Q–S = 13 days; P–R–S = 16 days. The critical path is P–R–S and the project duration is 16 days. The shorter path carries 16 − 13 = 3 days of total float.
2. $t_e = (5 + 4\times8 + 17) \div 6 = (5 + 32 + 17) \div 6 = 54 \div 6 = 9$ days. Variance = $[(17 - 5) \div 6]^2 = (12 \div 6)^2 = 4.0$.
3. A: ES 0, EF 3. B: ES 3, EF 7. C: ES 3, EF 8. D: ES 7, EF 9. E: ES 8, EF 14. F: ES 14 (it must wait for both D, finished day 9, and E, finished day 14), EF 17. Note how F's start is governed by E — the critical predecessor — not by D.
4. Z = (44 − 40) ÷ 2 = 2.0. Probability ≈ 97.7 %, as given — a firm promise, though never a certainty.
5. Total float is the slip an activity can absorb without delaying project completion; free float is the slip it can absorb without delaying the earliest start of any successor, and never exceeds total float. Critical activities have zero float, so their delay is project delay, day for day; floating activities have a buffer that management can spend deliberately.
6. Steps: confirm from the network which activities are critical; list crash options for critical activities only, with cost per day saved (cost slope); crash the cheapest first, re-checking the network after each step because the critical path may shift; stop when two weeks are recovered or when crash cost exceeds the benefit (penalty avoided / bonus gained). Risks: sharply rising cost for small time gains, quality and safety shortcuts under overtime pressure, worker fatigue, and a new path becoming critical and consuming the gain.